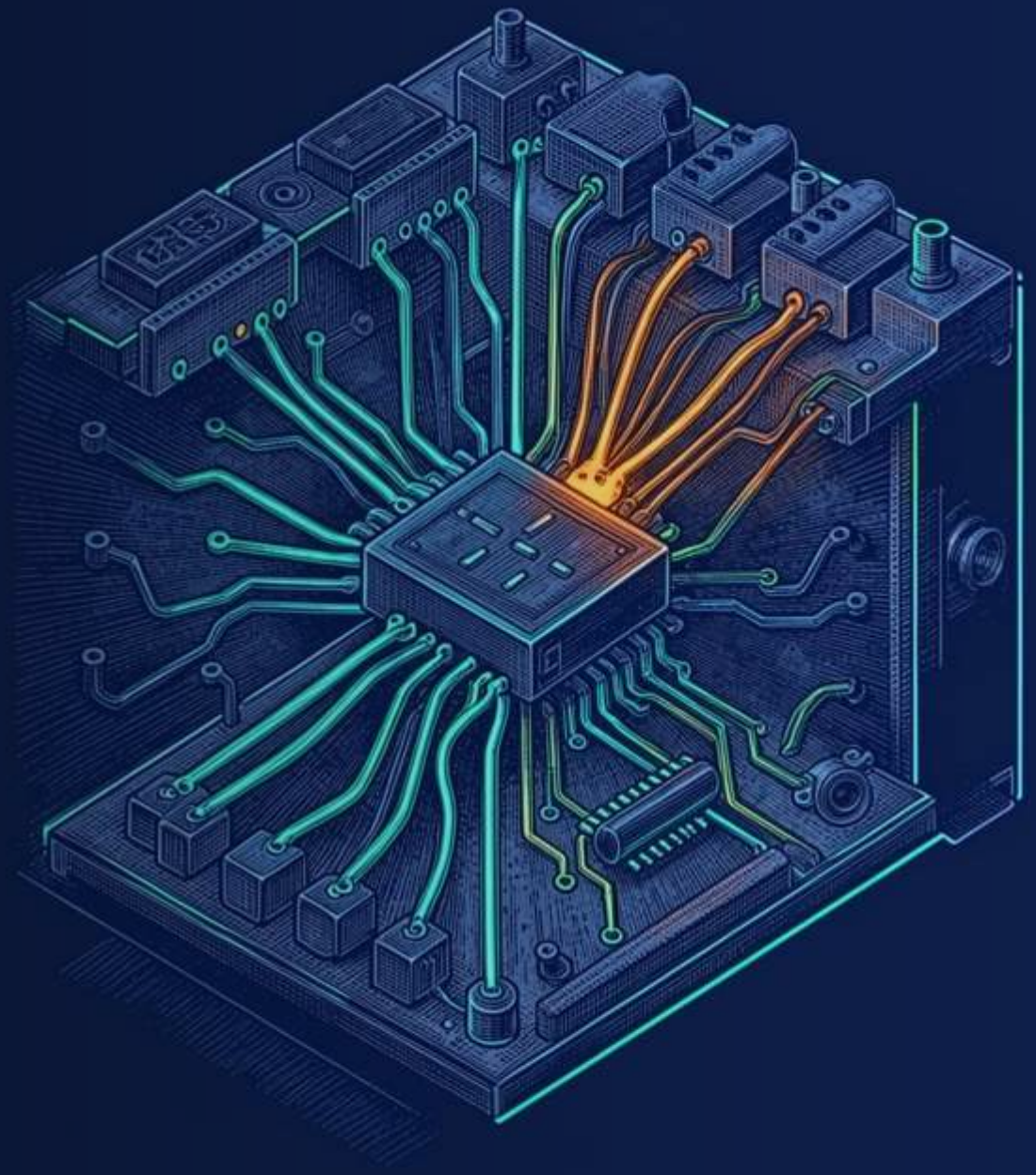
DIGITAL LOGIC · COMBINATIONAL CIRCUITS

8-to-3 Priority Encoder

Eight request lines enter. One winning input becomes a three-bit code.



THE CORE IDEA

One output code, eight possible requests

A priority encoder is a combinational circuit with 8 inputs and 3 binary outputs.

8	3	1
request inputs	encoded outputs	winning request

The circuit reports the binary index of the highest-priority active input. Unlike a basic encoder, it produces a defined result when several inputs are active.

PRIORITY RULE

Priority decides which input is seen

Assume active-high inputs Y7…Y0: Y7 has highest priority and Y0 lowest. X marks lower-priority inputs that no longer affect the result.

Input	A2	A1	A0	V
Y7	1	1	1	1
Y6, Y7=0	1	1	0	1
Y5, Y7…Y6=0	1	0	1	1
Y4, Y7…Y5=0	1	0	0	1
Y3, Y7…Y4=0	0	1	1	1
Y2, Y7…Y3=0	0	1	0	1
Y1, Y7…Y2=0	0	0	1	1
Y0, Y7…Y1=0	0	0	0	1
None	–	–	–	0

V = 1 when any input is active; V = 0 when all inputs are 0.

FROM PRIORITY TO LOGIC

Boolean equations turn priority into gates

Mask each lower input with the condition that every higher input is inactive.

MASKED WINNERS

$P7 = Y7$

$P6 = \neg Y7 \cdot Y6$

$P5 = \neg Y7 \cdot \neg Y6 \cdot Y5$

$P0 = \neg Y7 \cdot \neg Y6 \cdot \neg Y5 \cdot \neg Y4 \cdot \neg Y3 \cdot \neg Y2 \cdot \neg Y1 \cdot Y0$

OUTPUT OR-PLANES

$A2 = P7 + P6 + P5 + P4$

$A1 = P7 + P6 + P3 + P2$

$A0 = P7 + P5 + P3 + P1$

$V = Y7 + Y6 + Y5 + Y4 + Y3 + Y2 + Y1 + Y0$

NOT gates create the masks. AND gates identify one winning input. OR gates combine winners into A2A1A0 and V.

A LIVE REQUEST BUS

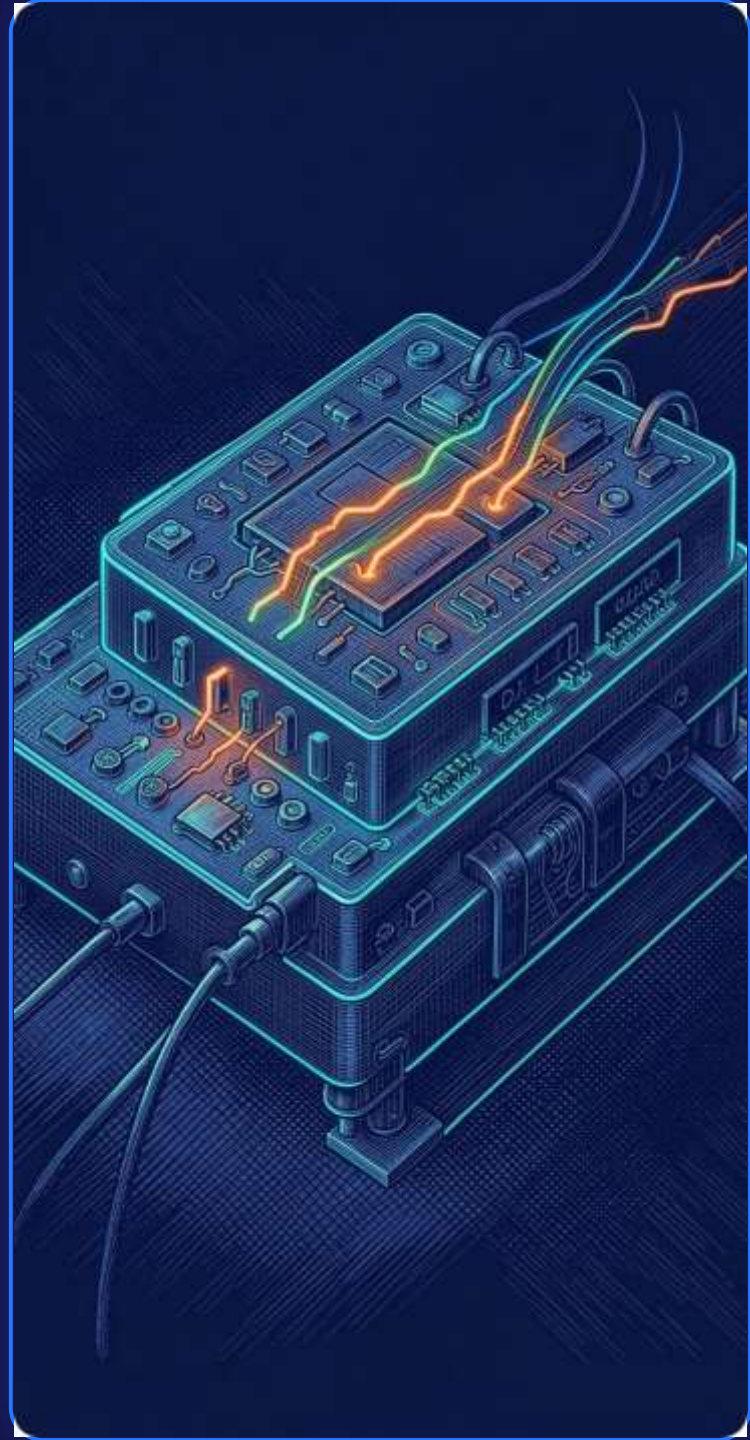
How the encoder works in real time

1. Monitor all eight request lines.
2. Block every input below the highest active request.
3. Pass the winning input's 3-bit binary code to the outputs.
4. Raise V so the next circuit knows the code is meaningful.

WORKED EXAMPLE

Y6 = 1 and Y2 = 1

Y6 wins, so A2A1A0=110 and V=1.



8 to 3 Priority Encoder (Logisim)

Overview

An 8 to 3 priority encoder has 8 input lines (I0 – I7) and 3 output lines (Y_2, Y_1, Y_0) along with a valid signal (V). If multiple inputs are high, the input with the highest priority (I7 highest, I0 lowest) is encoded.

If no input is high, V = 0 and the outputs are 000.

Logic Equations

$$Y_2 = I_7 + I_6 + I_5 + I_4$$

$$Y_1 = I_7 + I_6 + I_3 + I_2$$

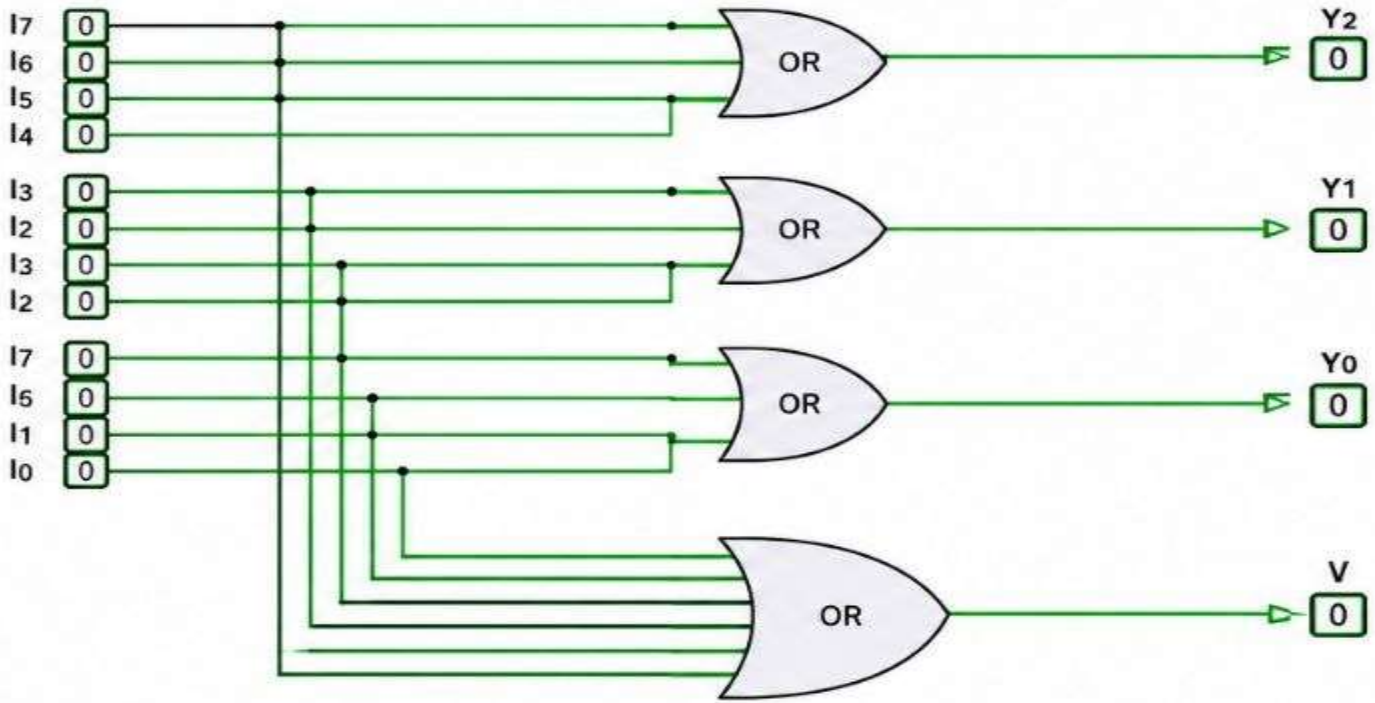
$$Y_0 = I_7 + I_5 + I_3 + I_1$$

$$V = I_7 + I_6 + I_5 + I_4 + I_3 + I_2 + I_1 + I_0$$

Priority Table (Behaviour)

Highest '1' Input	Y_2	Y_1	Y_0	V
I_7	1	1	1	1
I_6	1	1	0	1
I_5	1	0	1	1
I_4	1	0	0	1
I_3	0	1	1	1
I_2	0	1	0	1
I_1	0	0	1	1
I_0	0	0	0	1
None (all 0)	0	0	0	0

Logisim Circuit Diagram

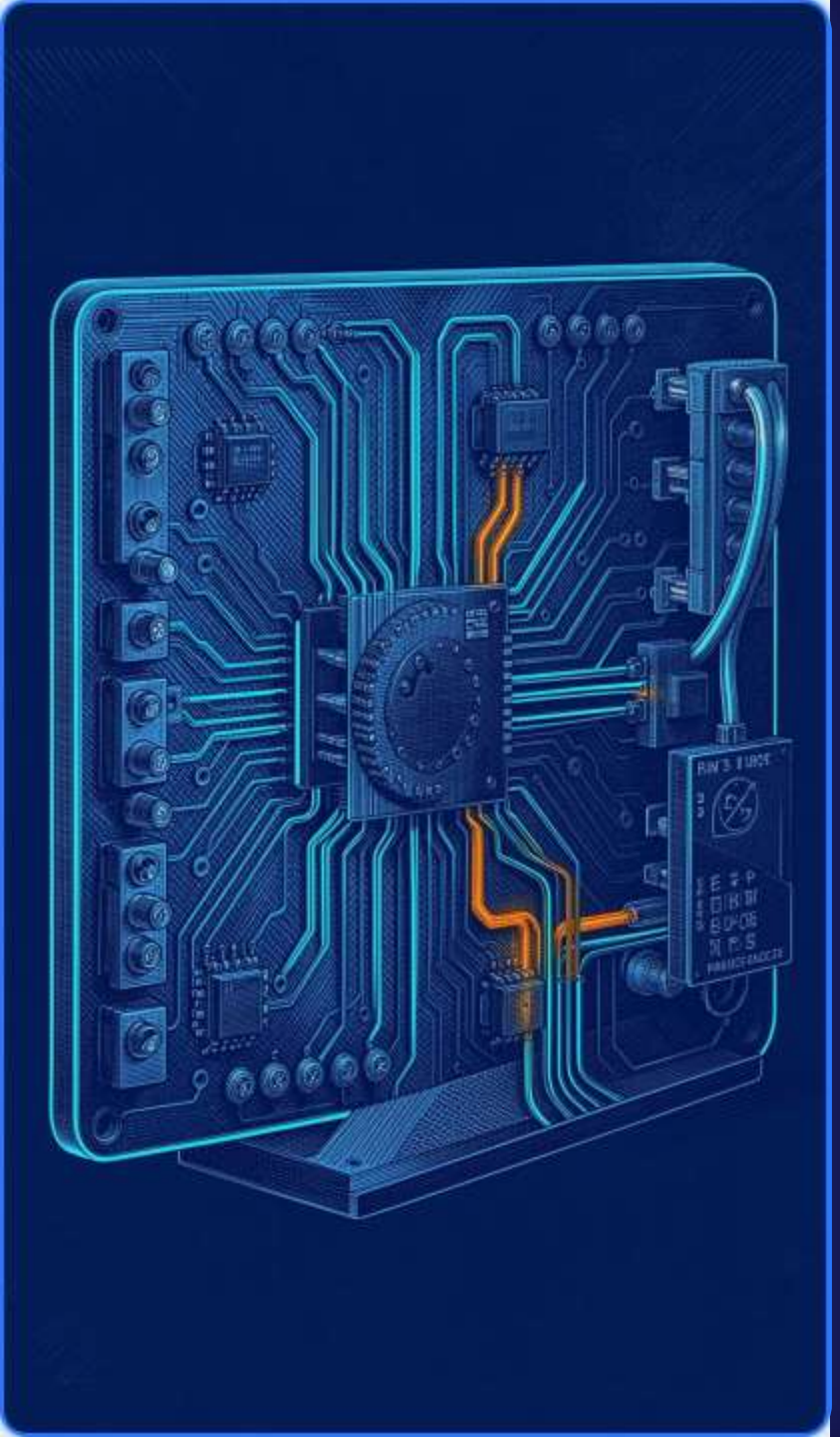


Components Used (in Logisim)

- OR gates (4-input, 8-input)
- Input pins (I0 – I7) → 8
- Output pins (Y2, Y1, Y0, V) → 4
- Wires and junctions

Notes

- I7 has the highest priority.
- If multiple inputs are high, the highest numbered input is encoded.
- When no input is high, V = 0 and outputs are 000.



WHY IT MATTERS

Where priority encoders are useful

Interrupt controllers

Service the most urgent request first.

Bus arbitration

Choose one requester when several devices compete.

Keyboard and switch scanning

Convert one active line into a compact code.

CPU and embedded control logic

Reduce many request lines to a small encoded interface.

Thank you

2620040029 - MAHITHA CHOWDARY

2620040030 - DHARANI KOLASANI

2620030695 - SUHAS REDDY

2620040031 - SYANTHANA PRIYA